

Anna University

MID-TERM EXAMINATION ? 2025

Machine Design ? Mid-Term Examination, 2025

Subject Code: ME501 Semester: 5 Regulation: R23

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Design for Static Loading)

1. Define Design for Static Loading. Explain its significance in Machine Design with a suitable example. [12 Marks]
2. Define Design for Static Loading. Explain its significance in Machine Design with a suitable example. [12 Marks]
3. List the advantages and limitations of Design for Static Loading. Explain any two in detail. [12 Marks]

SECTION B (Unit 2: Design for Fatigue Loading)

4. Explain Design for Fatigue Loading with a neat diagram/sketch and a suitable example. [12 Marks]
5. Write short notes on Design for Fatigue Loading. [8 Marks]
6. Describe the working principle of Design for Fatigue Loading with necessary equations. [8 Marks]

SECTION C (Unit 3: Shafts and Keys)

7. Write short notes on Shafts and Keys. [12 Marks]
8. Solve a problem related to Shafts and Keys. Show all the steps clearly. [8 Marks]
9. Explain Shafts and Keys with a neat diagram/sketch and a suitable example. [12 Marks]

SECTION D (Unit 4: Bearings and Lubrication)

10. How is Bearings and Lubrication implemented? Give an algorithm / procedure and explain. [12 Marks]
11. Describe the working principle of Bearings and Lubrication with necessary equations. [8 Marks]
12. Compare and contrast the key concepts involved in Bearings and Lubrication. [8 Marks]

SECTION E (Unit 5: Springs, Clutches and Brakes)

13. How is Springs, Clutches and Brakes implemented? Give an algorithm / procedure and explain. [12 Marks]
14. Define Springs, Clutches and Brakes. Explain its significance in Machine Design with

a suitable example. [12 Marks]

15. Solve a problem related to Springs, Clutches and Brakes. Show all the steps clearly.

[12 Marks]

*** End of Question Paper ***